

What is science?

A SYSTEM INVOLVING OBSERVATIONS AND TESTS USED TO FIGURE OUT THE MYSTERIES OF THE UNIVERSE AND EXPLAIN HOW NATURE WORKS

The word “science” means “knowledge” in Latin, and a scientist is someone who finds out new things. Scientific knowledge is the best way of describing the Universe—how it works and where it came from.

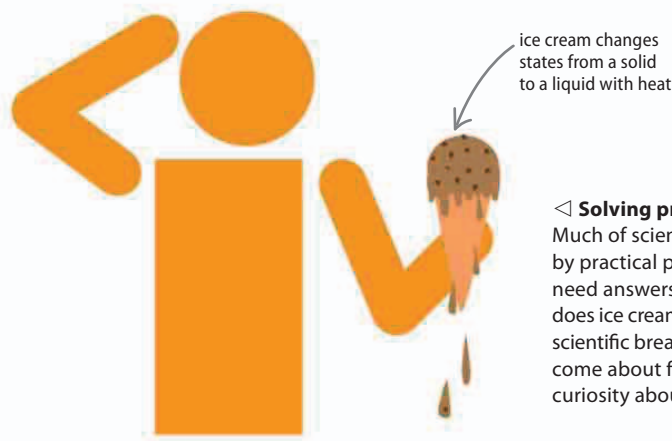
Science is...

...a collection of knowledge that is used to explain natural phenomena. The knowledge is arranged so that any fact can be confirmed by referring to other previously known facts.

...a way of uncovering new pieces of knowledge. This is achieved using a process of observation and testing that is designed to confirm whether a proposed explanation of something is true or false.

Answering questions

Science is an effective method of explaining natural phenomena. The way of doing this is known as the scientific method, which involves forming a theory about an unexplained phenomenon and doing an experiment to test it. Strictly speaking, the scientific method can only show whether a theory is false or not false. Once tested, a false theory is obviously no good and is discarded. However, a “not false” theory is the best explanation of a phenomenon we have—until, that is, another theory shows it to be false and replaces it.

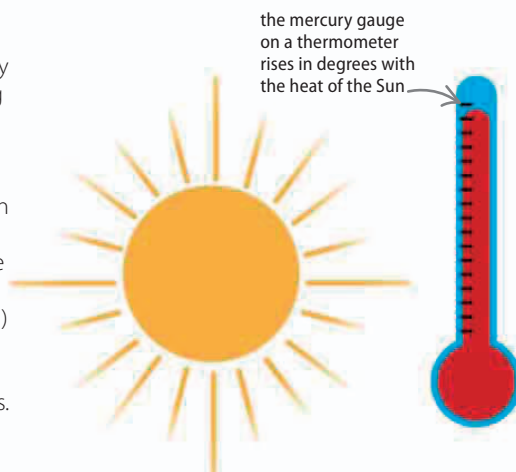


◁ Solving problems

Much of science is driven by practical problems that need answers, such as “Why does ice cream melt?” However, scientific breakthroughs also come about from pure curiosity about the Universe.

Measurements

Scientists need to make measurements as they gather evidence of how things behave. Saying a snake “was as long as an arm” is less useful than giving a precise length. Scientists use a system of measurements called the SI (Système International) units (see p.200), which include meters for length, kilograms for mass, seconds for time, and moles for measuring the quantity of a substance. All other units of measurement (eg, for force, pressure, or speed) are derived from the SI units. For this reason, metric units are given first throughout the book, with imperial equivalents in parentheses.



◁ Setting a scale

The degrees marked on a thermometer show the temperature rising and falling. However, like all units, the difference between one degree and the next is not something that is set by nature. The sizes of the units are generally set because they are practical to use in scientific calculations.